EPSR-test-ex61 clusters: overflow bin and no accumulation

Program(s): clusters | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `clusters`, the cluster analysis of EPSR [1]: two molecules are bonded if an atom1-atom2 pair lies within [rmin, rmax], bonded molecules form clusters, and the program writes the cluster-size histogram, the fraction of molecules in percolating clusters (spanning the box in at least one dimension) and the fraction in clusters of two or more molecules. This case checks the overflow bin and the default `nsumt`.

Code exercised

Fortran (original, fortran_src/):

- `clusters.f`: `run_clusters`, `find_clusters` (bond list, cluster labelling, percolation test)
- `read_and_save_files.f`: `read_input_file`, `outgofz`, `save_files_others`; `auxiliary_routines.f`: `r_int`
- `setup_symbols_generic.f`: `setup_symbols_clusters`

Rust (port, rust/epsr-core/src/):

- `tools/clusters.rs`: `run`, `find_clusters`, `outgofz`, `open_input_with`
- `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `CLUSTERS`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å^3), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [2]. Distributions: (1) water O1···O1 2.5-3.3 Å, the first-neighbour shell of the water network; (2) DMSO C···water O1 3.0-3.6 Å, methyl-group hydration. The reference gives percolating fractions 0.938 and 0.525 and fractions in clusters of 2 or more of 0.697 and 0.516.

Inputs

`oo.CLUSTERS.dat`: `nsize 10`, `ndist 2`, no `nsumt`. `200DMSO400Water.ato` copied unchanged from EPSR-test; the keyword file was written by hand (the Fortran program creates no template when the named file does not exist; keywords and defaults from `setup_symbols_generic.f`).

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.

`repeat 2`. `nsize 10` is smaller than the largest clusters, so the overflow branch is used (`gofz(nz,id)` collects all larger clusters; last row of the reference `.n01` non-zero). `nsumt` is not given (default -1), so no accumulation takes place (fc1 = 1) and nsumt stays -1 in the rewritten file. Compared after each pass: `oo.CLUSTERS.n01`, `oo.CLUSTERS.dat`, `fort.14` (written after unit 14 was closed, as in ex60).

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.